Lesson plan · for the teacher

Loops: repeat the pattern

Programming · Level 1 · Lesson 2

Time
40 min

Materials
Chalk for a grid; Arrow cards and number cards 1 to 4; Your phone (optional) with this lesson downloaded

Learning goals

- Spot repeated steps and turn them into a loop.
- Compare two programs by how many cards they need.
- Explain why a loop is easier to change.

Activity: Clap the loop

1. Warm-up (5 min): Clap “clap, clap, stamp” four times; pupils copy. Ask them to describe what you did in as few words as possible.
2. Teach (10 min): Write the twelve steps in full on the board, then “clap, clap, stamp: repeat 4 times”. Count the instructions in each. Ask what changes for ten times.
3. Create (10 min): Groups invent their own pattern of three or four moves and write it both ways, counting the instructions.
4. Perform (10 min): Each group performs its loop version. The class reads the full version and checks that nothing is missing.
5. Wrap-up (5 min): Ask where loops appear in songs, games and school routines. Then give the exit ticket.

Noisy but worth it: a pattern pupils perform is a loop they remember. No device is needed.

Discussion

- Why is a loop easier to change than the full version?
- Which loops do pupils already know from songs and games?
- When is a loop not worth it?

Exit ticket

- Write a pattern as a loop, and say how many steps it saves.

Try it in class tomorrow
Clap a pattern (clap, clap, stamp) and ask pupils to write it four times in full, then as “repeat 4 times”. Which is quicker to write, and to change to ten times?

Worksheet · for pupils

Loops: repeat the pattern

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Name _____ Class _____ Date _____

1 **"Repeat 3 times: forward 2." How many squares does the robot move?**

◯ 3 ◯ 5 ◯ 6

2 **Walk the robot to the flag in the opposite corner. Without repeats it needs 11 cards; with repeats you can use 6 at most.**

					★
↑					

Cut out cards (↑ forward, ↶ turn left, ↷ turn right) and lay out a program that takes the robot to the ★. Write it here:

3 **This is the robot's loop written in Python. How many times does the robot move forward?**

```
for i in range(4):
    forward()
```

◯ 3 ◯ 5 ◯ 4

4 **Match each everyday job to the loop that describes it.**

Sing the chorus three times	• • • •	For each pupil in the list
Call every name in the register	• • • •	Repeat until something happens
Stir the tea until the sugar dissolves	• • • •	Repeat 3 times

5 **A program with a loop can do exactly the same walk as a long program without one, with fewer cards.**

◯ True ◯ False

Answers · for the teacher

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1 **6**

Two squares, three times: 2 × 3 = 6. Pupils who answer 5 add the numbers instead of repeating the steps.

2 **Any program that takes the robot to the ★ with 6 cards or fewer.**

Forward × 4, forward × 1, turn right, forward × 4, forward × 1: five cards.

3 **4**

range(4) gives four values (0, 1, 2 and 3), so the indented line runs four times.

4 **Sing the chorus three times — Repeat 3 times**
Call every name in the register — For each pupil in the list
Stir the tea until the sugar dissolves — Repeat until something happens

Some loops run a set number of times, some once per item in a list, and some until a condition is met.

5 **True**

The loop does the same steps; it just says them once.

For you to think about

Which routine in your classroom is already a loop (the register, collecting books, a song)? Write it as "repeat … times", "for each …" or "until …".